ECEF系下的导航方程。

基本的导航方程：

$$a^{\gamma}_{ib} = f^{\gamma}_{ib} + \gamma^{\gamma}_{ib}$$

Let $\gamma = e$:

$$a^{e}_{ib} = f^{e}_{ib} + \gamma^{e}_{ib}$$

我们现在希望得到 v^e_{eb}，则有：

$$r^e_{eb} = r^e_{ib} - r^e_{ie} = r^e_{ib}$$

同时求导： $\dot{r}^e_{eb} = v^e_{eb} = \dot{r}^e_{ib}$ 仍=号

再求导： $\dot{v}^e_{eb} = \ddot{r}^e_{ib} \neq a^e_{ib}$ （解析系和参考系之间有旋转）

$$r^e_{ib} = C^e_i r^i_{ib}$$

$$\Rightarrow \dot{r}^e_{ib} = \dot{C}^e_i r^i_{ib} + C^e_i \dot{r}^i_{ib}$$

$$= -\Omega^e_{ie} C^e_i r^i_{ib} + C^e_i \dot{r}^i_{ib}$$

$$\Rightarrow \ddot{r}^e_{ib} = -\dot{\Omega}^e_{ie} C^e_i r^i_{ib} - \Omega^e_{ie} \dot{C}^e_i r^i_{ib} - \Omega^e_{ie} C^e_i \dot{r}^i_{ib} + \dot{C}^e_i \dot{r}^i_{ib} + C^e_i \ddot{r}^i_{ib}$$

$$= \Omega^e_{ie} \Omega^e_{ie} C^e_i r^i_{ib} - 2\Omega^e_{ie} C^e_i \dot{r}^i_{ib} + C^e_i \ddot{r}^i_{ib}$$

$$= \Omega^e_{ie} \Omega^e_{ie} C^e_i r^i_{ib} - 2\Omega^e_{ie} C^e_i \dot{r}^i_{ib} + a^e_{ib}$$

$v^e_{ib} = C^e_i \dot{r}^i_{ib} \neq \dot{r}^e_{ib}$

Let $\dot{r}^e_{ib} = (C^e_i r^i_{ib})^{\cdot} = \dot{C}^e_i r^i_{ib} + C^e_i \dot{r}^i_{ib}$

$$= -\Omega^e_{ie} C^e_i r^i_{ib} + C^e_i \dot{r}^i_{ib}$$

$$\Rightarrow C^e_i \dot{r}^i_{ib} = \dot{r}^e_{ib} + \Omega^e_{ie} C^e_i r^i_{ib}$$

代入得： $\dot{v}^e_{eb} = \ddot{r}^e_{ib} = \Omega^e_{ie} \Omega^e_{ie} r^e_{ib} - 2\Omega^e_{ie}(\dot{r}^e_{ib} + \Omega^e_{ie} r^e_{ib}) + a^e_{ib}$

$$= -\Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e - 2\Omega_{ie}^e \dot{\underline{r}}_{ib}^e + \underline{a}_{ib}^e$$

$$= -\Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e - 2\Omega_{ie}^e v_{eb}^e + a_{ib}^e$$

再根据 $\underline{a}_{ib}^e = C_b^e \underline{f}_{ib}^b + \underline{\gamma}_{ib}^e$

替换掉 a_{ib}^e :

$$\dot{\underline{v}}_{eb}^e = C_b^e f_{ib}^b + \underline{\gamma}_{ib}^e - \Omega_{ie}^e \Omega_{ie}^e \underline{r}_{ib}^e - 2\Omega_{ie}^e \underline{v}_{eb}^e$$

$$= C_b^e f_{ib}^b + \underline{g}_b^e - 2\Omega_{ie}^e \underline{v}_{eb}^e$$

ECEF 系的方程.

$$\dot{\underline{v}}_{eb}^e = -\Omega\Omega r - 2\Omega v + a$$